UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

# Introduction to Partial Differentiation

Complete study pack for class 001: functions of several variables, limits and continuity of a function of two variables, the definition and notation of partial derivatives, and second-order and mixed partial derivatives

Compiled from the PESU class 001 slide deck, its two AV-summary transcripts, the Unit 1 class-work sheet and past ESA papers. Generated 15 October 2026.

# · How to use this pack

Class 001 has one deck and two recorded lectures: “Introduction to Partial Differentiation” (topics 1–2) and “Continuity and differentiability of a function of two independent variables” (topics 3–5). Topics 1–3 are the background (functions of two variables, limits, continuity) and are never asked by themselves in the ESA; topics 4–5 (computing partial derivatives and mixed partials) are used in every Unit 1 question. The geometrical interpretation and the problems on higher-order partials announced at the end of the second lecture are class 002, a separate pack. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

| | |
|---|---|
| ESA ★★★ | Asked repeatedly in past ESA papers |
| ESA ★★ | Asked in past ESA papers or computed in class-work problems |
| ESA ★ | Conceptual: taught in class, not examined so far |
| ADDED FOR CLARITY | Standard context that is not in the class material |
| TRANSCRIPT ONLY | Said in the lecture but not on any slide |
| VIDEO ONLY | Seen only in a frame of the class video |
| NOT ON CURRENT SLIDES | From older material or past papers, not this year's slides |
| S001·15 | Deck 001, slide 15 (rendered .pptx) |
| T001a 00:14:20 | Transcript “Introduction to Partial Differentiation”, segment start |
| T001b 00:11:41 | Transcript “Continuity and differentiability of a function of two independent variables” |
| CW U1 P7 | Unit 1 class-work problem 7 |
| ESA Dec-23 1a | Past paper: Dec 2023, question 1.a |

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6. Appendix A · Symbol glossary
7. Appendix B · Sources, coverage and gaps

# 1 Functions of two or more variables

Most quantities depend on more than one variable: the area of a rectangle on its length and breadth, the volume of a cylinder on $r$ and $h$, a potential on $x, y, z, t$. A function of two variables $z = f(x, y)$ assigns to each ordered pair $(x, y)$ in a domain $D$ a unique real number. Its graph is a surface in three dimensions, so tools are used to plot it; this unit works with two variables, and everything extends to $n$ variables.

### Key terms and definitions

**Function of two variables.** A function of two variables denoted by $z = f(x, y)$ is a rule that assigns to each ordered pair of real numbers $(x, y)$ in a set $D$ a unique real number denoted by $f(x, y)$. The set $D$ is the domain of $f$ and its range is the set of values that $f$ takes on: $\{f(x, y) : (x, y) \in D\}$.
S001·8 · T001a 00:05:46

**Independent and dependent variables.** In $z = f(x, y)$, $x$ and $y$ are independent variables and $z$ is the dependent variable (a bivariate function); with three independent variables one writes $w = f(x, y, z)$, and in general $f(x_1, x_2, \ldots, x_n)$. (transcript only) T001a 00:02:18 · T001a 00:02:54

## Graphs and diagrams

$$z = \sin x \cos y$$

$$-\pi \le x,\ y \le \pi;\ \ -1 \le z \le 1$$

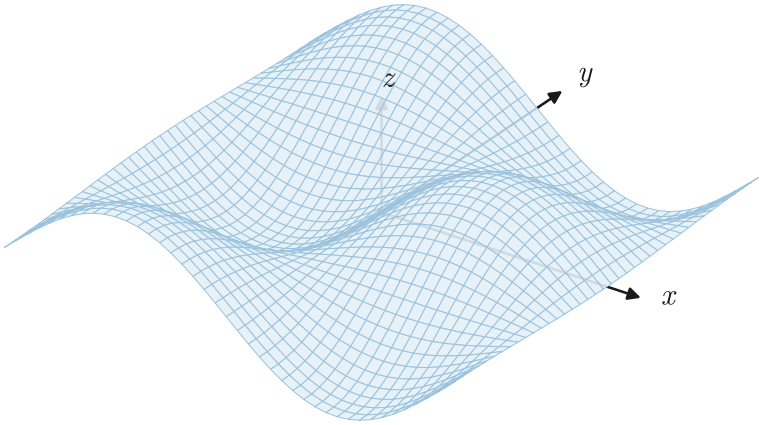



**Figure G1.1.** Graph of $z = \sin x \cos y$. ESA ★

S001·8 · T001a 00:09:07

***How to read it.*** A function of two variables is a surface over the $xy$-plane: above each point $(x, y)$ the height is $f(x, y)$. $\sin x \cos y$ is an egg-box of hills and hollows between $-1$ and $1$; the slide's Gnuplot picture shows the same surface.

$z = e^{1 - x^2 - y^2}$

top $(0, 0, e)$

green: level curves $z =$ const (circles)

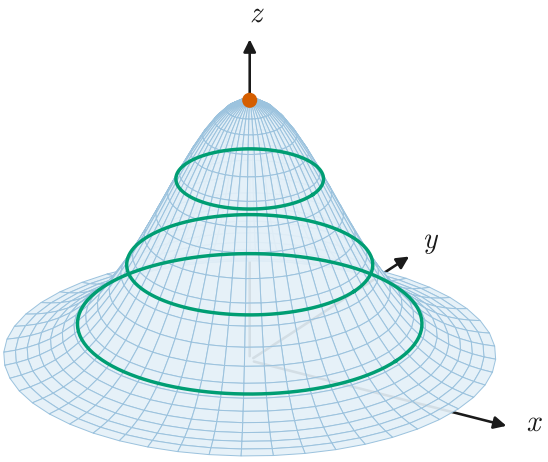



**Figure G1.2.** Graph of $z = e^{1-x^2-y^2}$. ESA ★

S001·8 · T001a 00:09:07

***How to read it.*** A single bell-shaped hill with its top $z = e$ above the origin, falling to 0 in every direction. The green level curves (drawn on the surface) are circles $x^2 + y^2 =$ constant.

## Important points and why

1. Examples on the slides: area of a rectangle $A(l, b)$; volume of a cylinder $v(r, h)$; electric potential in a cathode-ray tube $V = f(x, y, z, t)$; propagation of an electromagnetic wave (magnetic field, electric field, wavelength, direction); weather (topography, pressure, humidity, temperature); spread of a pandemic (area, population, awareness, immunity, age and gender distribution). ESA ★ S001·4 · S001·5 · S001·6 · S001·7 · T001a 00:03:26
2. A single-variable function is enough to understand a concept, but to draw a conclusion about a real situation we almost always need several variables. ESA ★ TRANSCRIPT ONLY T001a 00:05:16
3. $f$ maps an ordered pair to one real number: for $z = x^3y$, the pair $(2, 1)$ is mapped to 8. ESA ★ T001a 00:07:16
4. Graph of $z = f(x, y)$: a surface, which needs three axes; a function of three variables would need four dimensions, so the unit sticks to two variables. The slide shows $\sin x \cos y$ and $e^{1-x^2-y^2}$ (Figures G1.1–G1.2). ESA ★ S001·8 · T001a 00:08:35 · T001a 00:09:07

SHORT-NOTE RECAP

Quantities depending on several variables need functions of several variables (area $A(l,b)$, volume $v(r,h)$, potential $V(x,y,z,t)$, weather, pandemics). $z=f(x,y)$ assigns to each $(x,y)\in D$ a unique real number; $D$ is the domain, the set of values the range. The graph is a surface.

# 2 Limit of a function of two variables

In one variable $x$ can approach $a$ only from the left or the right. In the plane $(x, y)$ can approach $(a, b)$ along infinitely many paths: lines, parabolas, ellipses. The limit is $L$ only if every path gives $L$. Proving a limit exists is hard; proving it does not exist is easy: find two paths that give different values.

### Key terms and definitions

**Limit of a function of two variables.** A function $f(x, y)$ is said to approach the limit $L$ as $(x, y)$ approaches $(a, b)$, if the limit remains the same along any path of $(x, y)$ approaching $(a, b)$: $\lim_{(x,y)\to(a,b)} f(x, y) = L$. S001·9 · T001a 00:11:01

### Formulas

**E2.1 · Limit of f(x, y)** ESA ★

$$\lim_{(x,y)\to(a,b)} f(x, y) = L$$

| | |
|---|---|
| gives | The value $f$ approaches near $(a, b)$. |
| applies | The same $L$ along every path to $(a, b)$. |
| note | For a polynomial (or any function continuous at the point) just substitute. |

S001·9 · T001a 00:13:59

**E2.2 · Two-path test (limit does not exist)** ESA ★

$$\lim_{\text{path 1}} f \neq \lim_{\text{path 2}} f \implies \lim_{(x,y)\to(a,b)} f(x, y) \text{ does not exist}$$

| | |
|---|---|
| gives | A proof that the limit does not exist. |
| applies | Substitution gives $\frac{0}{0}$. |
| note | Lines $y = mx$ through the origin are the usual first try: if the answer depends on $m$, the limit does not exist. |

S001·10 · T001a 00:17:42

## Graphs and diagrams

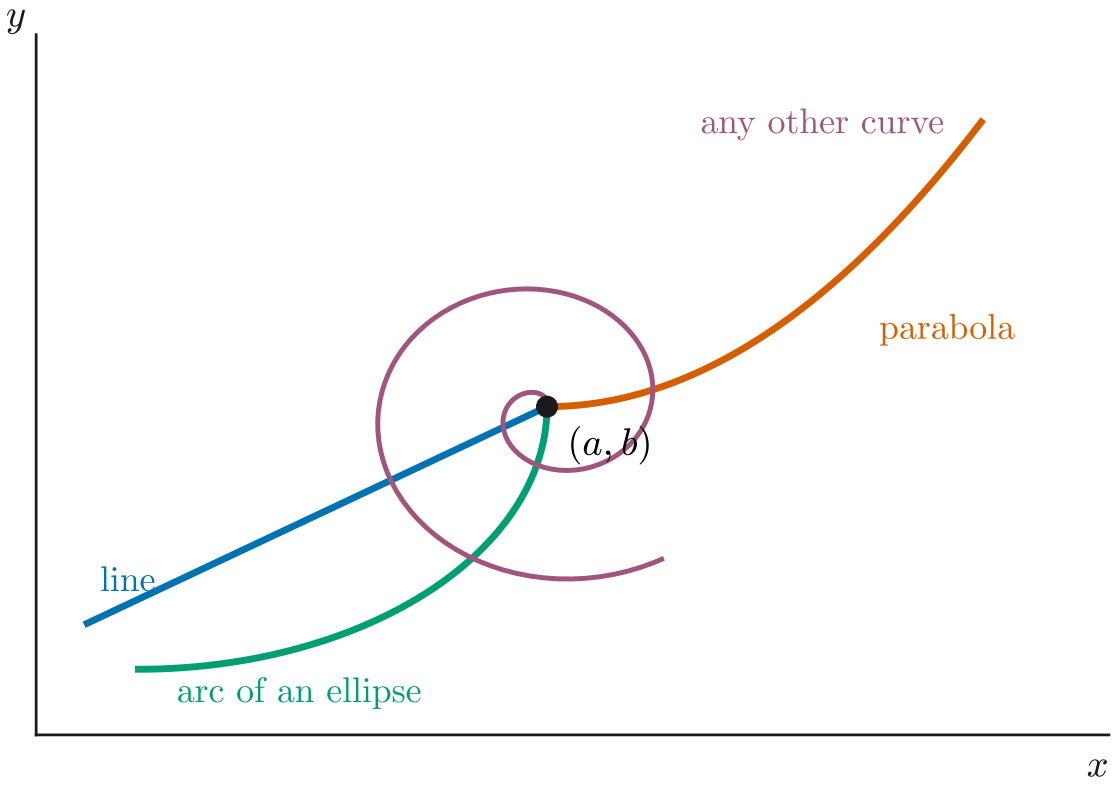



**Figure G2.1.** Paths along which $(x, y)$ can approach $(a, b)$. ESA ★

S001·9 · T001a 00:11:52 · T001a 00:12:19

***How to read it.*** In one variable $x$ reaches $a$ only from the left or the right. In the plane, $(x, y)$ can reach $(a, b)$ along a line, a parabola, an arc of an ellipse, or any other path. The limit exists only if all of them give the same value.

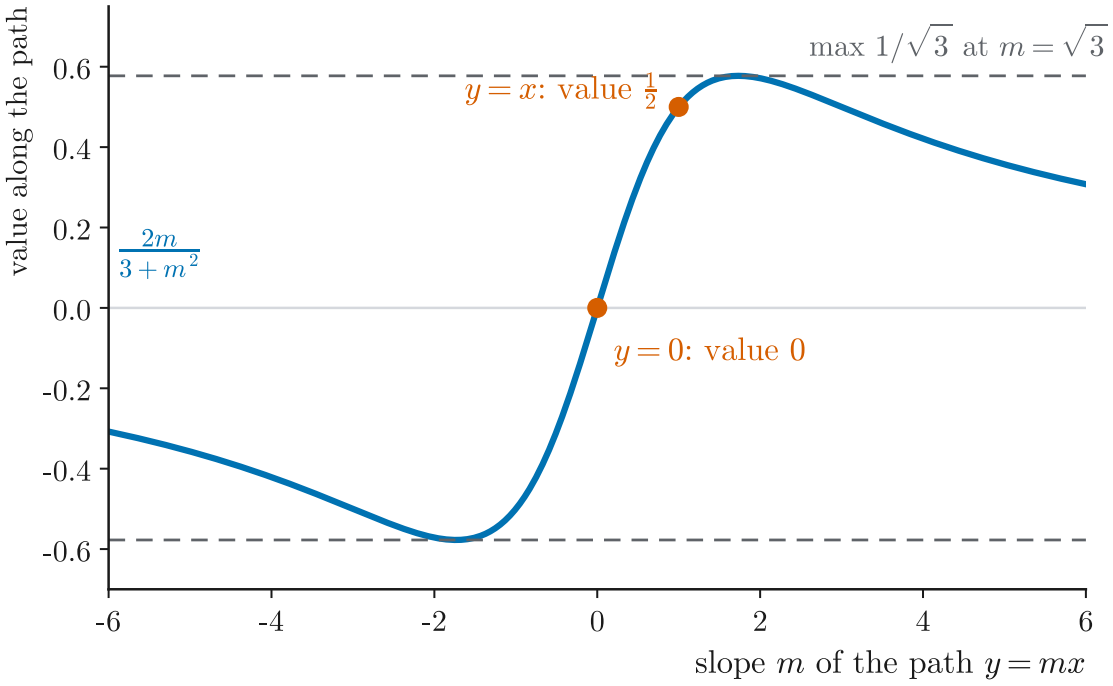
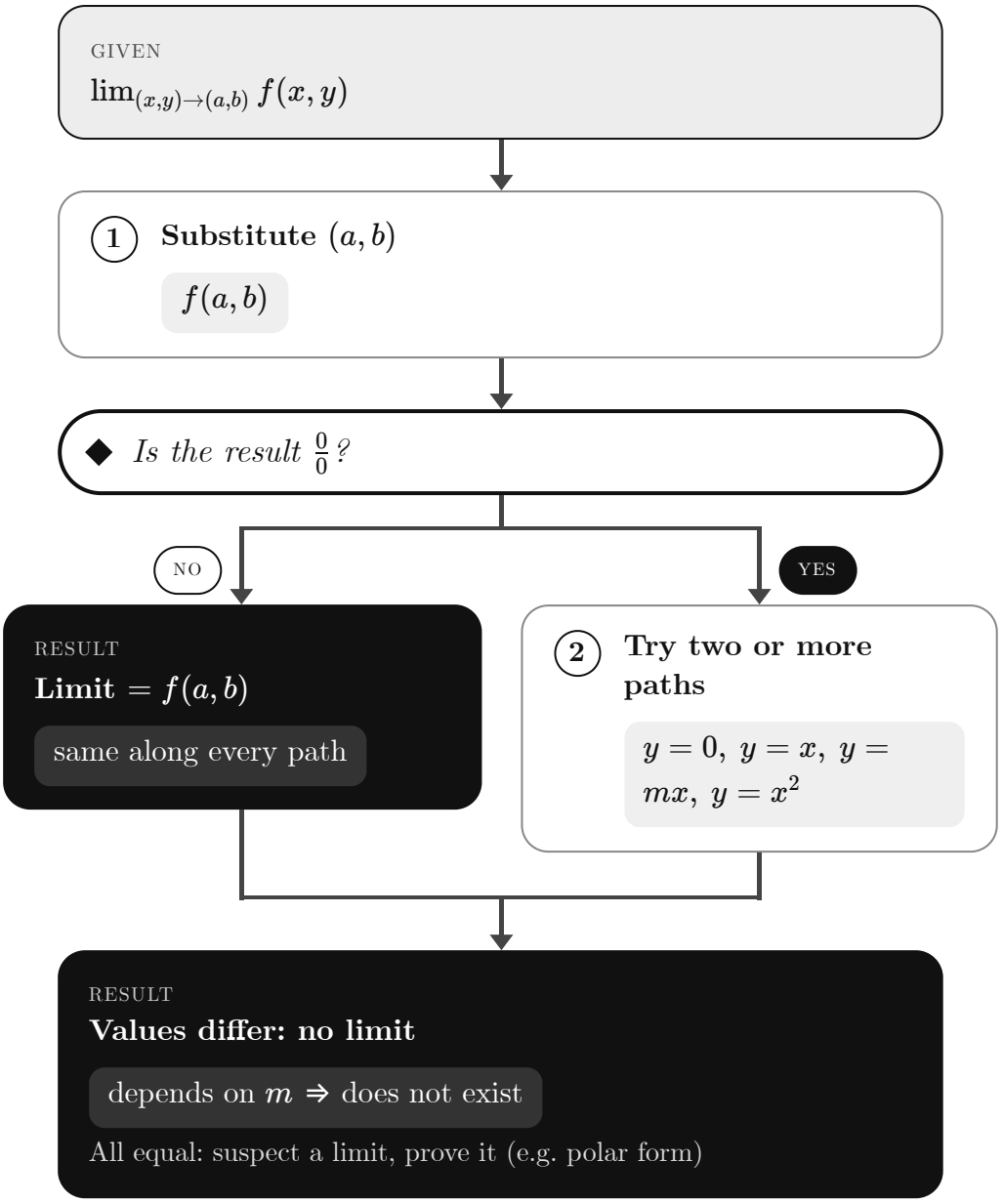



**Figure G2.2.** $\frac{2xy}{3x^2+y^2}$ along the line $y=mx$. ESA ★★

ADDED FOR CLARITY S001·10 · T001a 00:17:24

***How to read it.*** The value of $\frac{2xy}{3x^2+y^2}$ on the line $y=mx$ is the constant $\frac{2m}{3+m^2}$, plotted against the slope $m$. $m=0$ gives 0, $m=1$ gives $\frac{1}{2}$, and the largest value is $\frac{1}{\sqrt{3}}$ at $m=\sqrt{3}$. Because the value changes with the path, the limit at the origin does not exist.

## Flowcharts




***Remark.*** Agreement along a few paths is only evidence; disagreement is proof.

***Exam trap.*** Read the point carefully: slide $(2,1)$ gives $-8$, the lecture's $(2,-1)$ gives $-6$.

**Figure F2.1.** Finding (or disproving) a limit at a point ESA ★★

S001·9 · S001·10 · T001a 00:13:59 · T001a 00:17:42

## One variable and two variables

| | $y = f(x)$ | $z = f(x, y)$ |
|---|---|---|
| **Domain** | Real numbers | Ordered pairs $(x, y)$ |
| **Graph** | A curve in the plane | A surface in space |
| **Approach to the point** | From the left or the right | Along infinitely many paths |
| **Limit exists if** | Left and right limits agree | Every path gives the same value |
| **Continuous if** | $\lim_{x \to c} f = f(c)$ | $\lim_{(x,y) \to (a,b)} f = f(a, b)$ |
| **Derivative** | $\frac{dy}{dx}$ | $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$ (others constant) |

S001·9 · S001·11 · T001a 00:09:54 · T001a 00:11:52

## Important points and why

1. One variable: $x \to a$ from the left or the right only. Two variables: $(x, y) \to (a, b)$ along any of infinitely many paths (Figure G2.1). ESA ★ S001·9 · T001a 00:11:32 · T001a 00:12:34
2. If substitution gives a number (a polynomial, no $\frac{0}{0}$), that number is the limit, whatever the path. ESA ★ T001a 00:13:59
3. It is hard to show that a limit exists but easy to show it does not: two paths with different values are enough. ESA ★★ S001·10 · T001a 00:17:42
4. Agreement along a few paths proves nothing: for $\frac{2xy}{3x^2+y^2}$, the $x$-axis and the parabola $y = x^2$ both give 0, but $y = x$ gives $\frac{1}{2}$. ESA ★★ T001a 00:16:48 · T001a 00:17:24

**Why can one path never prove that a limit exists?**
There are infinitely many paths, and a function can agree on any finite number of them and still differ on another: $\frac{2xy}{3x^2+y^2}$ gives 0 on the $x$-axis and on $y = x^2$ but $\frac{1}{2}$ on $y = x$. To prove existence one needs an estimate for all paths at once (e.g. polar coordinates: $\frac{3x^2y}{x^2+y^2} = 3r\cos^2\theta\sin\theta \to 0$).
ADDED FOR CLARITY T001a 00:17:42 · S001·12

***Exam trap.*** Same value along two or three paths does not prove that the limit exists; different values along two paths does prove it does not. T001a 00:16:48 · T001a 00:17:42

***Exam trap.*** Slide 001·9 uses the point $(2, 1)$ (limit $-8$); the lecture uses $(2, -1)$ (limit $-6$). Read the point in the question. ADDED FOR CLARITY S001·9 · T001a 00:12:59

## Worked examples

### W2.1 · Limit of a polynomial (slide 001·9) ESA ★

**GIVEN** Evaluate $\lim_{(x,y)\to(2,1)}(x^2 - 2xy + 3y^2 - 4x + 3y - 6)$; the lecture also does the point $(2, -1)$.

**STEP 1** A polynomial is continuous everywhere, so substitute.

**STEP 2** At $(2,1)$: $4 - 4 + 3 - 8 + 3 - 6 = -8$.

**STEP 3** At $(2,-1)$: $4 + 4 + 3 - 8 - 3 - 6 = -6$.

**ANSWER** $-8$ at $(2,1)$; $-6$ at $(2,-1)$

**CHECK** sympy agrees for both points. The value is reached along every path.

S001·9 · T001a 00:12:59 · T001a 00:13:47

### W2.2 · Show that a limit does not exist (slide 001·10) ESA ★★

**GIVEN** Show that $\lim_{(x,y)\to(0,0)} \dfrac{2xy}{3x^2 + y^2}$ does not exist.

**STEP 1** Substitution gives $\frac{0}{0}$: indeterminate.

**STEP 2** Along $y = 0$ ($x$-axis): $\frac{0}{3x^2} = 0 \to 0$.

**STEP 3** Along $y = x^2$: $\frac{2x^3}{3x^2+x^4} = \frac{2x}{3+x^2} \to 0$.

**STEP 4** Along $y = x$: $\frac{2x^2}{4x^2} = \frac{1}{2}$.

**STEP 5** General line $y = mx$: $\frac{2m}{3+m^2}$, which depends on $m$ (Figure G2.2).

**ANSWER** Different paths give $0$ and $\frac{1}{2}$: the limit does not exist

**CHECK** sympy: $\frac{2m}{m^2+3}$ along $y = mx$.

S001·10 · T001a 00:14:20 · T001a 00:15:04 · T001a 00:17:03

SHORT-NOTE RECAP

$\lim_{(x,y)\to(a,b)} f = L$ if every path to $(a,b)$ gives $L$. Polynomial: substitute. $\frac{0}{0}$: try paths ($y = 0$, $y = x$, $y = mx$, $y = x^2$); two different values ⇒ no limit. $\frac{2xy}{3x^2+y^2}$: $0$ on $y = 0$, $\frac{1}{2}$ on $y = x$.

# 3 Continuity of a function of two variables

$f$ is continuous at $(a,b)$ when $f(a,b)$ is defined, the limit exists, and $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$, as in one variable. Geometrically the surface has no hole or break there. A function can fail to be continuous even though the limit exists ($\frac{3x^2y}{x^2+y^2}$ at the origin), or because the limit does not exist ($\frac{2xy}{3x^2+y^2}$).

### Key terms and definitions

**Continuity at a point.** $f(x,y)$ is continuous at $(a,b)$ if $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$: $f(a,b)$ must be defined, the limit must exist, and the two must be equal. S001·11 · T001b 00:00:38 · T001a 00:18:46

> *"A function is continuous at a point if the graph doesn't have any holes or breaks at that point. Graphing the function is helpful. If a function is not continuous at $(a,b)$, this does not mean the limit does not exist. We will need to consider the approaching $(a,b)$ from different paths."* S001·11

### Formulas

**E3.1 · Continuity at (a, b)** ESA ★

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$$

| | |
|---|---|
| gives | The test for continuity. |
| applies | Three conditions: $f(a,b)$ defined, limit exists, both equal. |
| note | One variable: $\lim_{x\to c} f(x) = f(c)$. |

S001·11 · T001b 00:00:38

## Graphs and diagrams

$$z = \frac{3x^2 y}{x^2 + y^2}$$

limit 0 at the origin; $f(0,0)$ undefined

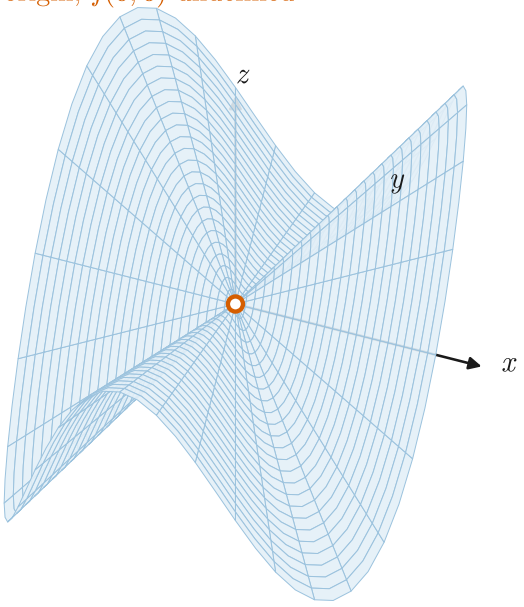



**Figure G3.1.** $z = \frac{3x^2y}{x^2+y^2}$: the limit at the origin exists. ESA ★

S001·12 · T001b 00:02:41

***How to read it.*** The surface folds near the origin but every direction leads to height 0 there (vermilion point). It is missing only the single point $(0,0)$, where $f$ is undefined: a removable discontinuity.

$$z = \frac{2xy}{3x^2 + y^2}$$

ray $y = x$: height $\frac{1}{2}$

$x$-axis: height $0$

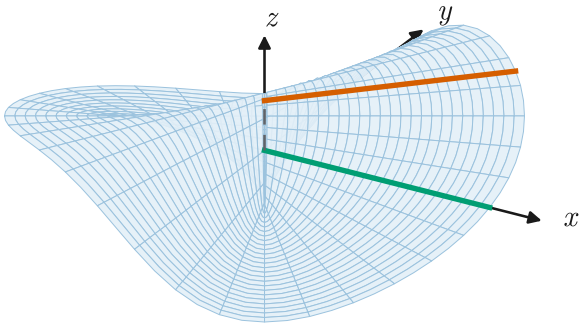
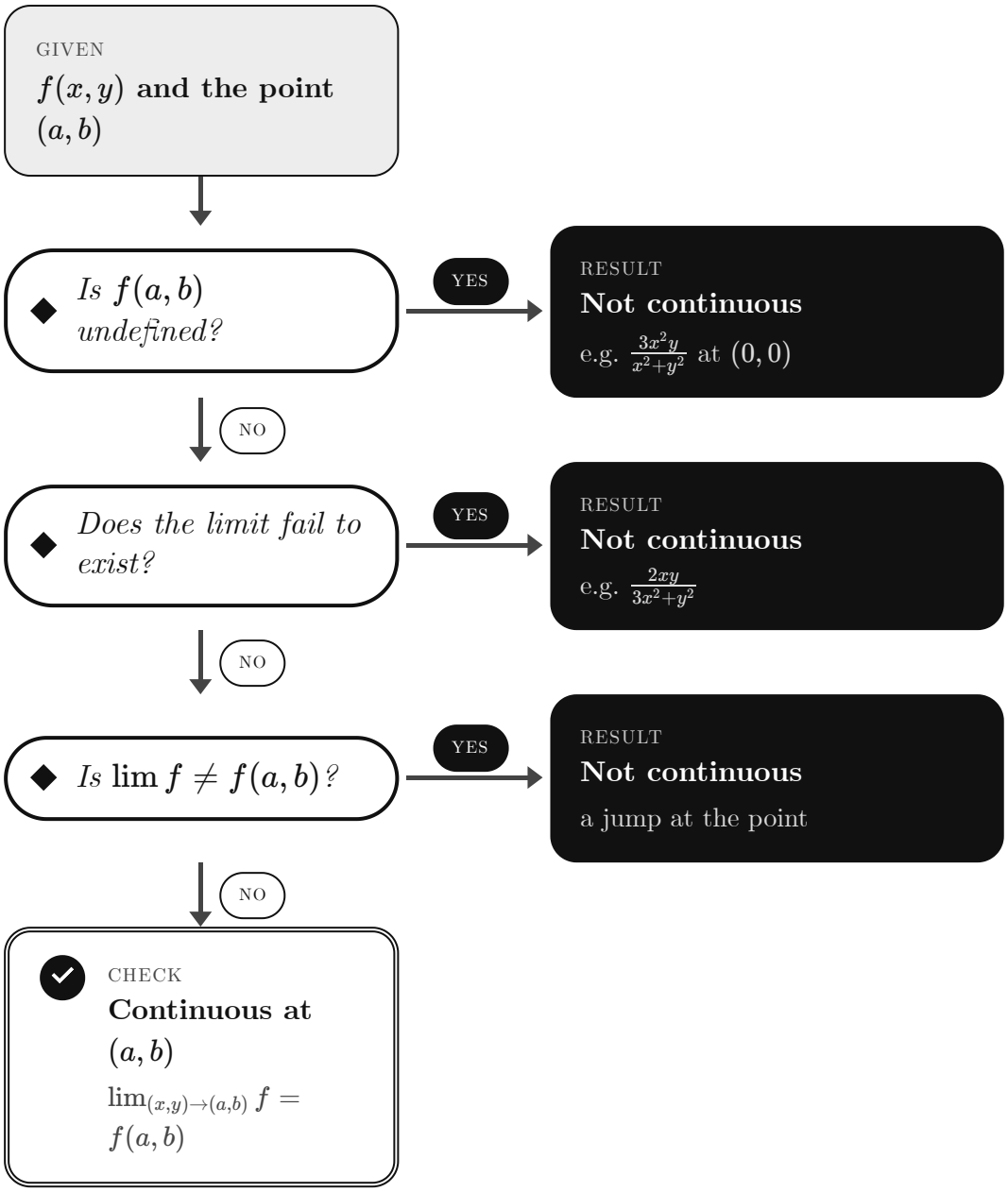



**Figure G3.2.** $z = \frac{2xy}{3x^2+y^2}$: the limit at the origin does not exist. ESA ★

S001·13 · T001b 00:06:13

***How to read it.*** Each ray from the origin keeps a constant height $\frac{2m}{3+m^2}$, so the surface is a twisted fan of straight lines meeting at the $z$-axis at different heights (vermilion: the ray $y = x$ at height $\frac{1}{2}$; blue: the $x$-axis at height 0). There is no single value at the origin.

## Flowcharts




***Remark.*** Only the first rung failing with the limit existing is a removable discontinuity: define $f(a,b)$ as the limit.

**Figure F3.1.** Is f continuous at (a, b)? ESA ★

S001·11 · T001a 00:18:46 · T001b 00:03:17

## Important points and why

1. One variable: $f$ is continuous at $c$ if (i) $\lim_{x\to c} f(x)$ exists, (ii) $f(c)$ exists, (iii) they are equal. The same three conditions define continuity for $f(x,y)$. ESA ★ T001a 00:18:46 · S001·11

2. $\frac{3x^2y}{x^2+y^2}$: the limit at the origin exists (it is 0) but $f(0,0)$ is $\frac{0}{0}$, so $f$ is not continuous there. $\frac{2xy}{3x^2+y^2}$: the limit does not exist and $f(0,0)$ is undefined (Figures G3.1–G3.2). ESA ★ S001·12 · S001·13 · T001b 00:04:14 · T001b 00:05:44
3. To decide continuity in higher dimensions, graph the function: a hole, fold or sudden drop in the surface shows a discontinuity. ESA ★ S001·11 · T001b 00:06:45 · T001a 00:18:07

**Why is $\frac{3x^2y}{x^2+y^2}$ not continuous at the origin even though its limit is 0?**
Continuity needs $f(0,0)$ to be defined and equal to the limit. Here $f(0,0)$ is $\frac{0}{0}$, so the third condition cannot be checked. Defining $f(0,0)=0$ would make it continuous.
ADDED FOR CLARITY S001·12 · T001b 00:04:14

***Exam trap.*** "Not continuous" does not mean "the limit does not exist" (slide 001·11): $\frac{3x^2y}{x^2+y^2}$ has limit 0 at the origin. S001·11 · T001b 00:02:10

***Exam trap.*** Continuous is not the same as smooth: $z=|x|+|y|$ is continuous everywhere but has a corner at the origin. The lecture's "continuous if the surface is smooth" is loose.
ADDED FOR CLARITY T001b 00:06:26

## Worked examples

### W3.1 · Limit exists, function not continuous (slide 001·12) ESA ★

**GIVEN** Discuss the continuity of $f(x,y)=\dfrac{3x^2y}{x^2+y^2}$ at the origin.

**STEP 1** $f(0,0)=\frac{0}{0}$ is not defined, so condition (ii) fails: $f$ is not continuous at $(0,0)$.

**STEP 2** The limit exists: in polar form $x=r\cos\theta$, $y=r\sin\theta$, $f=3r\cos^2\theta\sin\theta$, and $|f|\le 3r\to 0$ for every path (added for clarity).

**STEP 3** So $\lim_{(x,y)\to(0,0)} f=0$.

**ANSWER** Not continuous at $(0,0)$; the limit is $0$ (removable: set $f(0,0)=0$)

**CHECK** sympy: polar form $3r\sin\theta\cos^2\theta$.

S001·12 · T001b 00:02:41 · T001b 00:04:27

### W3.2 · Limit does not exist, function not continuous (slide 001·13) ESA ★

**GIVEN** Discuss the continuity of $f(x,y)=\dfrac{2xy}{3x^2+y^2}$ at the origin.

**STEP 1** The limit does not exist (W2.2).

**STEP 2** $f(0,0)$ is also undefined.

**ANSWER** Not continuous at $(0,0)$; no choice of $f(0,0)$ can make it continuous

**CHECK** Figure G3.2 shows the jump along different directions.

S001·13 · T001b 00:05:05 · T001b 00:05:44

SHORT-NOTE RECAP

Continuous at $(a,b)$: $f(a,b)$ defined, limit exists, $\lim f = f(a,b)$; no hole or break in the surface.
Not continuous ≠ no limit: $\frac{3x^2y}{x^2+y^2}$ has limit 0 but $f(0,0)$ undefined.

# 4 Partial derivatives: history, definition and notation

A partial derivative is the derivative with respect to one variable with the others held constant. The idea is already in Leibniz's product rule; the symbol $\partial$ came from Legendre (1786) and was reintroduced by Jacobi (1841). $\frac{\partial z}{\partial x} = z_x = f_x$ is defined by a limit, and every rule of ordinary differentiation (product, quotient, chain) still holds.

### Key terms and definitions

**Partial derivative.** A partial derivative of a function of several variables is the derivative with respect to one of those variables, with the others held constant. S001·15 · T001b 00:12:52

### Formulas

**E4.1 · Partial derivatives from the definition** ESA ★★

$$\frac{\partial z}{\partial x} = \lim_{h \to 0} \frac{f(x+h, y) - f(x, y)}{h}$$
$$\frac{\partial z}{\partial y} = \lim_{h \to 0} \frac{f(x, y+h) - f(x, y)}{h}$$

| | |
|---|---|
| gives | $z_x$ (y held constant) and $z_y$ (x held constant). |
| applies | When the limits exist. |
| note | Only one variable is moved by $h$; the other stays fixed. |

S001·15

**E4.2 · Notation for first-order partials** ESA ★★

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial x} = z_x = f_x, \qquad \frac{\partial z}{\partial y} = \frac{\partial f}{\partial y} = z_y = f_y$$

| | |
|---|---|
| gives | The equivalent symbols. |
| applies | $z = f(x, y)$. |
| note | The symbol is $\partial$ (“partial”, often read “dou”), not $\delta$ or $d$. |

S001·15 · T001b 00:11:41 · T001b 00:13:05

## Flowcharts

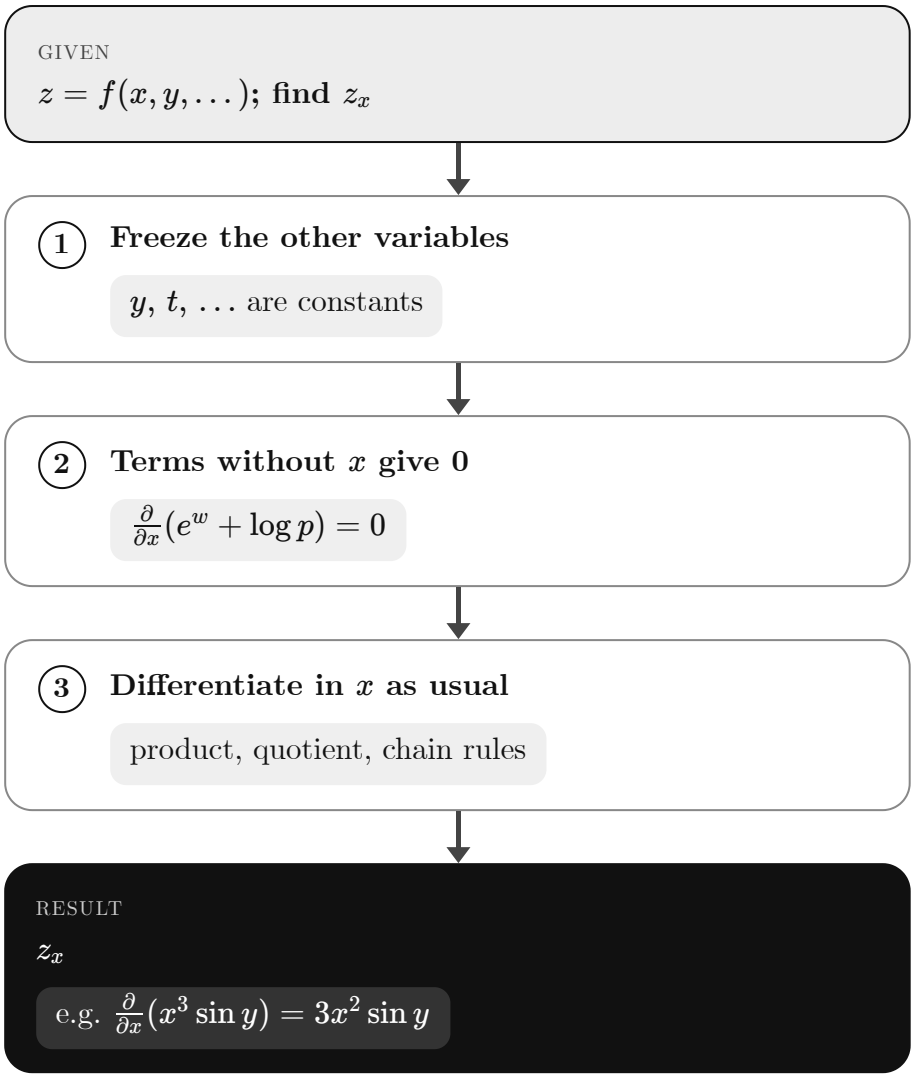



**Figure F4.1.** Computing a partial derivative ESA ★★

S001·15 · T001b 00:15:08 · T001b 00:13:57

## Important points and why

1. The product rule is already partial differentiation: in $\frac{d}{dx}(x^3 \sin x) = x^3 \cos x + 3x^2 \sin x$, each term treats the other factor as a constant $k$. Leibniz: $d(uv) = u\,dv + v\,du$. ESA ★ S001·14 · T001b 00:07:56 · T001b 00:10:18
2. History: Leibniz first used partial differentiation (product rule); Adrien-Marie Legendre created the modern symbol $\partial$ in 1786 but later abandoned it; Carl Gustav Jacob Jacobi reintroduced it in 1841. ESA ★ S001·14 · T001b 00:11:23
3. To find $z_x$, look only at the terms containing $x$; everything else is a constant, however many other variables there are. ESA ★★ T001b 00:15:08 · T001b 00:15:48
4. A function of $x, y, t$ has three first partials $z_x$, $z_y$, $z_t$; for $z_x$ treat $y$ and $t$ as constants. ESA ★ T001b 00:13:20
5. All rules of ordinary differentiation hold for partial derivatives: product, quotient, chain rule, implicit and composite functions. ESA ★★ T001b 00:13:57 · T001b 00:14:16

**Why does every rule of ordinary differentiation still work?**
With the other variables frozen, $f(x, y)$ is an ordinary function of $x$ alone, and $\frac{\partial f}{\partial x}$ is its ordinary derivative (E4.1 is the usual limit with $y$ fixed).

T001b 00:13:57 · S001·15

***Exam trap.*** The partial symbol is $\partial$, not $\delta$ (the lecture stresses this) and not $d$. T001b 00:11:41

## Worked examples

### W4.1 · A partial derivative from the definition — ESA ★

**given** Use E4.1 to find $f_x$ for $f(x, y) = x^2 y$.

**step 1** $\frac{f(x+h,y)-f(x,y)}{h} = \frac{(x+h)^2 y - x^2 y}{h} = \frac{(2xh+h^2)y}{h} = (2x + h)y.$

**step 2** Let $h \to 0$: $2xy$.

**answer** $\boxed{f_x = 2xy}$

**check** Same as differentiating $x^2$ with $y$ constant; sympy agrees.

added for clarity S001·15

### W4.2 · Partials of a function of several variables (lecture) — ESA ★★

**given** $z = x^3 \sin y + e^w + \log p$ (a function of $x, y, w, p$). Find $z_x$, $z_p$, $z_y$ and $z_{xy}$.

**step 1** $z_x$: only $x^3 \sin y$ contains $x$; $\sin y$ is a constant: $3x^2 \sin y$.

**step 2** $z_p$: only $\log p$ contains $p$: $\frac{1}{p}$.

**step 3** $z_y$: only $\sin y$ varies: $x^3 \cos y$.

**step 4** $z_{xy}$: differentiate $z_x = 3x^2 \sin y$ with respect to $y$: $3x^2 \cos y$.

**answer** $\boxed{z_x = 3x^2 \sin y,\ z_p = \dfrac{1}{p},\ z_y = x^3 \cos y,\ z_{xy} = 3x^2 \cos y}$

**check** sympy agrees; $z_{yx} = \frac{\partial}{\partial x}(x^3 \cos y) = 3x^2 \cos y$ too (Clairaut).

T001b 00:14:40 · T001b 00:15:48 · T001b 00:16:26 · T001b 00:16:44

short-note recap

Partial derivative: derivative in one variable, others constant. $z_x = \lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h}$; notation $\frac{\partial z}{\partial x} = z_x = f_x$. Leibniz (product rule), Legendre 1786, Jacobi 1841. All differentiation rules hold.

# 5 Second-order and mixed partial derivatives

Differentiating again gives $z_{xx}$, $z_{yy}$ and the mixed partials $z_{xy}$, $z_{yx}$. Clairaut's theorem: if the second derivatives are continuous, the order of differentiation does not matter, $z_{xy} = z_{yx}$. Subscripts are read in the order of differentiation; in the $\partial$ notation the order is read from right to left.

### Key terms and definitions

**Mixed (crossed) partial derivatives.** $z_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)$ and $z_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right)$: differentiate with respect to one variable, then the other. S001·16 · T001b 00:18:24

> *"Note 1: The crossed or mixed partial derivatives are in general equal (Clairaut's theorem): $\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)$. That is, the order of differentiation is immaterial if the second order derivatives involved are continuous."* S001·17 · T001b 00:19:57

> *"Note 2: In the subscript notation, the subscripts are written in same order in which the differentiation is carried out, while in the $\partial$ notation the order is opposite. For example, $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = f_{xy}$."* S001·17

### Formulas

**E5.1 · Second-order partials** ESA ★★

$$z_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial x^2} = f_{xx}, \qquad z_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial y^2} = f_{yy}$$

| | |
|---|---|
| GIVES | Second derivatives in one variable. |
| APPLIES | Differentiate $z_x$ again in $x$ (or $z_y$ in $y$). |
| NOTE | Same rules as ordinary second derivatives. |

S001·16 · T001b 00:17:04

**E5.2 · Mixed partials** ESA ★★

$$z_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial y\,\partial x} = f_{xy}, \qquad z_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial x\,\partial y} = f_{yx}$$

| | |
|---|---|
| GIVES | The two crossed derivatives. |
| APPLIES | Differentiate in one variable, then the other. |
| NOTE | Subscripts: left to right. $\partial$ notation: right to left. |

S001·16 · S001·17 · T001b 00:18:41

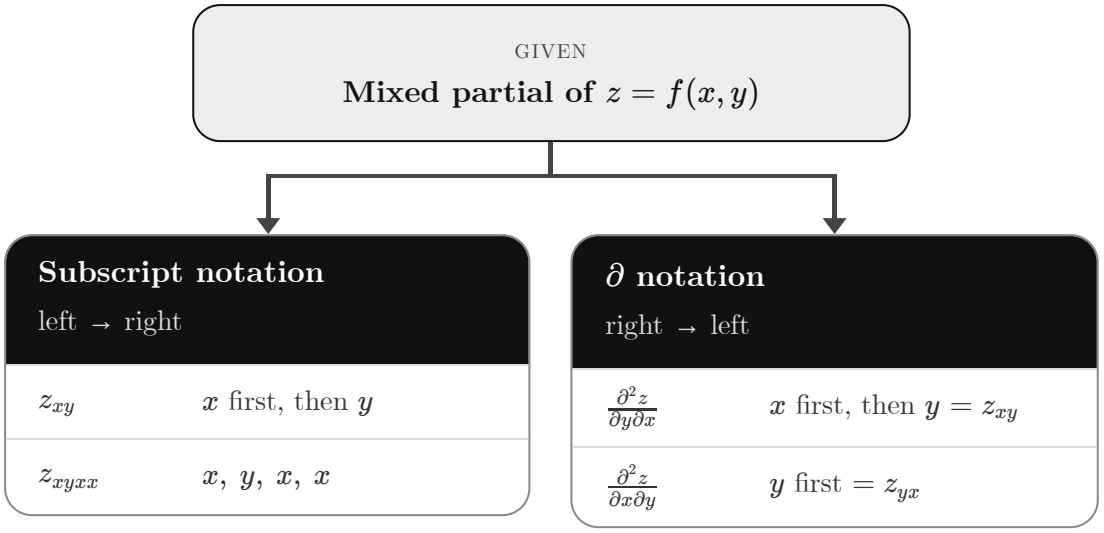
### E5.3 · Clairaut's theorem ESA ★★★

$$f_{xy} = f_{yx} \qquad \left(\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 f}{\partial x\,\partial y}\right)$$

| | |
|---|---|
| GIVES | The order of differentiation does not matter. |
| APPLIES | $f_{xy}$ and $f_{yx}$ continuous near the point. |
| NOTE | Called "a very important theorem" in the lecture; it is why Taylor's series has $2hkf_{xy}$. |

S001·17 · T001b 00:19:57 · T001b 00:20:10

## Flowcharts




***Remark.*** By Clairaut's theorem the order rarely changes the answer, but the notation is still examined.

**Figure F5.1.** Reading the order of a mixed partial ESA ★★

S001·17 · T001b 00:18:51

## Important points and why

1. Read $z_{xyxx}$ left to right: $z \to z_x \to z_{xy} \to z_{xyx} \to z_{xyxx}$ (differentiate in $x$, then $y$, then $x$, then $x$). ESA ★ T001b 00:18:51 · T001b 00:19:21
2. Clairaut's theorem is "a very important theorem": whenever the second partials are continuous (true for all examination functions away from bad points), compute whichever mixed partial is easier. ESA ★★★ S001·17 · T001b 00:20:10
3. Verifying $f_{xy} = f_{yx}$ is a standard class-work question: $x^y$ (CW P2) and $x^2 \tan^{-1}\frac{y}{x} - y^2 \tan^{-1}\frac{x}{y}$ (CW P8, W5.3). ESA ★★ CW U1 P2 · CW U1 P8

***Exam trap.*** $\frac{\partial^2 z}{\partial y\,\partial x}$ means $x$ first, then $y$: it is $z_{xy}$, not $z_{yx}$. S001·17

***Exam trap.*** Clairaut needs continuous second partials: for $f = \frac{xy(x^2-y^2)}{x^2+y^2}$, $f(0,0)=0$, one gets $f_{xy}(0,0) = -1$ but $f_{yx}(0,0) = 1$. ADDED FOR CLARITY S001·17

## Worked examples

### W5.1 · Second-order partials of a function of three variables

ESA ★★

**GIVEN** If $f(x,y,z) = e^{x^2+y^2+z^2}$, find $f_{xx}$, $f_{yy}$, $f_{zz}$ at $(-1,1,-1)$.

**STEP 1** $f_x = 2xe^{x^2+y^2+z^2}$.

**STEP 2** $f_{xx} = (2+4x^2)e^{x^2+y^2+z^2}$ (product rule).

**STEP 3** At $(-1,1,-1)$: $x^2+y^2+z^2 = 3$, so $f_{xx} = 6e^3$; by symmetry $f_{yy} = (2+4y^2)e^3 = 6e^3$, $f_{zz} = 6e^3$.

**ANSWER** $\boxed{f_{xx} = f_{yy} = f_{zz} = 6e^3}$

**CHECK** CW answer agrees; sympy agrees.

CW U1 P4

### W5.2 · Find n so that v satisfies an equation

ESA ★★

**GIVEN** Find $n$ so that $v = r^n(3\cos^2\theta - 1)$ satisfies $\dfrac{\partial}{\partial r}\left(r^2\dfrac{\partial v}{\partial r}\right) + \dfrac{1}{\sin\theta}\dfrac{\partial}{\partial\theta}\left(\sin\theta\dfrac{\partial v}{\partial\theta}\right) = 0$.

**STEP 1** $v_r = nr^{n-1}(3\cos^2\theta - 1)$, so $\frac{\partial}{\partial r}(r^2 v_r) = n(n+1)r^n(3\cos^2\theta - 1)$.

**STEP 2** $v_\theta = -6r^n\cos\theta\sin\theta$, so $\sin\theta\, v_\theta = -6r^n\cos\theta\sin^2\theta$.

**STEP 3** $\frac{\partial}{\partial\theta}(\cdot) = -6r^n(2\sin\theta\cos^2\theta - \sin^3\theta)$; divide by $\sin\theta$: $-6r^n(3\cos^2\theta - 1)$.

**STEP 4** Sum: $[n(n+1) - 6]\, r^n(3\cos^2\theta - 1) = 0 \Rightarrow n^2 + n - 6 = 0$.

**ANSWER** $\boxed{n = 2 \text{ or } n = -3}$

**CHECK** CW answer $(-3,\ 2)$ agrees; sympy solves to $[-3, 2]$. Same pattern as ESA Jan-24 s4 1c (worked in the class 002 pack).

CW U1 P7 · ESA Jan-24 s4 1c

### W5.3 · Verify that the mixed partials are equal ESA ★★

**GIVEN** If $u = x^2 \tan^{-1}\dfrac{y}{x} - y^2 \tan^{-1}\dfrac{x}{y}$, show that $\dfrac{\partial^2 u}{\partial x \partial y} = \dfrac{\partial^2 u}{\partial y \partial x} = \dfrac{x^2 - y^2}{x^2 + y^2}$.

**STEP 1** $u_x = 2x \tan^{-1}\frac{y}{x} + x^2 \cdot \frac{-y}{x^2+y^2} - y^2 \cdot \frac{y}{x^2+y^2} = 2x \tan^{-1}\frac{y}{x} - y$.

**STEP 2** $u_{xy} = \frac{\partial}{\partial y}(u_x) = \frac{2x^2}{x^2+y^2} - 1 = \frac{x^2-y^2}{x^2+y^2}$.

**STEP 3** Similarly $u_y = x - 2y \tan^{-1}\frac{x}{y}$, and $u_{yx} = 1 - \frac{2y^2}{x^2+y^2} = \frac{x^2-y^2}{x^2+y^2}$.

**ANSWER** $\boxed{u_{xy} = u_{yx} = \dfrac{x^2 - y^2}{x^2 + y^2}}$

**CHECK** sympy: both simplify to $\frac{x^2-y^2}{x^2+y^2}$.

CW U1 P8

SHORT-NOTE RECAP

$z_{xx}$, $z_{yy}$, $z_{xy} = \frac{\partial}{\partial y}(z_x) = \frac{\partial^2 z}{\partial y \partial x}$, $z_{yx}$. Clairaut: $z_{xy} = z_{yx}$ when the second partials are continuous. Subscripts left to right, $\partial$ right to left.

# A Symbol glossary

| Symbol | Name | Meaning | Appears in |
|---|---|---|---|
| $z = f(x, y)$ | Function of two variables | $x, y$ independent, $z$ dependent | T1 |
| $D$ | Domain | Set of ordered pairs where $f$ is defined | T1 |
| $L$ | Limit | Common value along every path | E2.1 |
| $(a, b)$ | Point of approach | Where the limit or continuity is tested | E2.1, E3.1 |
| $m$ | Slope of a path | Path $y = mx$ in the two-path test | E2.2, W2.2 |
| $\partial$ | Partial (“dou”) | Partial derivative sign; not $\delta$, not $d$ | E4.1–E5.3 |
| $h$ | Increment | Small change in one variable in the definition | E4.1 |
| $z_x$, $f_x$ | First partial in $x$ | $\frac{\partial z}{\partial x}$ with $y$ constant | E4.2 |
| $z_{xx}$, $z_{yy}$ | Second partials | $\frac{\partial^2 z}{\partial x^2}$, $\frac{\partial^2 z}{\partial y^2}$ | E5.1 |
| $z_{xy}$ | Mixed partial | $x$ first, then $y$; $= \frac{\partial^2 z}{\partial y \partial x}$ | E5.2, E5.3 |

# B Sources, coverage and gaps

| Source | Path | Use |
|---|---|---|
| Deck 001 (.pptx rendered; .slides.md; .slides.txt) | pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.* | 18 slides; content on 3–17 |
| Transcript 001a (all 4 formats) | pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Introduction_to_Partial_Differentiation__7d6d5767ec830adb.* | 19.7 min: motivation, definition, limits |
| Transcript 001b (all 4 formats) | pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Continuity_and_differentiability_of_a_function_of_two_independent_variable__543f4d65e9b6ed52.* | 21.1 min: continuity, history, definition, mixed partials |
| Unit 1 & 2 class-work sheet (UE24MA141A) | pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf | Problems 1–8 (with answers) |
| ESA papers, UE22MA141A and UE23MA141A (8 papers) | pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf | No direct question; two partial-derivative computations (class 002) |
| Videos 001a, 001b (.mp4) | pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_*.mp4 | Not opened: every slide is readable and the board work is read out in the transcripts |

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. No question asks for a limit, continuity or the definition of a partial derivative. Computing partial derivatives is asked directly twice: Dec 2023 Q1.a (third-order partials of $e^x \log y + \cos y \log x$) and Jan 2024 set 4 Q1.c (find $n$ for $\theta = t^n e^{-r^2/4t}$). Both are worked in the class 002 pack; this pack works the class-work problems of the same type (CW P4, P7, P8). Every other Unit 1 question (chain rule, Euler, Taylor, maxima and minima, Lagrange) uses the rules of this session.

### Known gaps, conflicts and corrections

1. Slide 001·9 asks for $\lim_{(x,y)\to(2,1)}(x^2 - 2xy + 3y^2 - 4x + 3y - 6)$, which is $-8$. The lecture (T001a 00:12:59) works the point $(2, -1)$ instead and gets $-6$. Both values are correct for their points (sympy); W2.1 gives both.
2. Slide 001·12's text is about $f = \frac{3x^2 y}{x^2+y^2}$ (whose limit at the origin is 0), but the Gnuplot caption on the slide's picture reads $\frac{3xy}{y^2+x^2}$, a different function whose limit at the origin does not exist (along $y = mx$ it is $\frac{3m}{1+m^2}$). Figure G3.1 redraws the function in the text.
3. Slide 001·12 says the function is not continuous at $(0, 0)$ "as $f(0, 0)$ is not defined". Added for clarity: the limit is 0, so defining $f(0, 0) = 0$ removes the discontinuity (a removable discontinuity); with $2xy/(3x^2 + y^2)$ (slide 001·13) no value of $f(0, 0)$ helps.

4. The lecture says a function is continuous "if the surface is smooth" (T001b 00:06:26). Continuity only needs no holes or jumps; a continuous surface can have a sharp edge or corner ($z = |x| + |y|$). Corrected as an exam trap.
5. Slide 001·17 says mixed partials "are in general equal". Precisely, Clairaut's theorem needs $f_{xy}$ and $f_{yx}$ continuous near the point; the standard counter-example $\frac{xy(x^2-y^2)}{x^2+y^2}$ has $f_{xy}(0,0) = -1 \neq f_{yx}(0,0) = 1$ (added for clarity; sympy).
6. The lecture asks the class to prove Clairaut's theorem (T001b 00:20:10) and says the proof is short. The proof needs the mean value theorem as well as the limit definition; it is not on the slides and is not reproduced here.
7. The lecture's history (T001b 00:10:53) says Leibniz used other letters for partial derivatives in a letter to L'Hôpital. The letters are not on the slide and could not be confirmed from the material; only the slide's facts (Leibniz, Legendre 1786, Jacobi 1841) are kept.
8. The lecture's example function (T001a, T001b 00:15:16) adds a term in $t$ that the recognition renders unclearly; W4.2 keeps the clearly spoken terms $x^3 \sin y + e^w + \log p$, which show the same point.
9. The lecture opens with "Unit Two" (T001a 00:00:01); partial differentiation is Unit 1 of UE26MA141A.
10. The deck's motivation slides (001·5–7) carry stock images (a cathode-ray tube, a weather chart, COVID-19 posters); they are described in words, not redrawn.
11. Transcript recognition errors were corrected silently: "Clydoids" → Clairaut's, "legendary" → Legendre, "Lobithal" → L'Hôpital, "Ebo" → above, "parable" → parabola, "cathode ratio" → cathode-ray tube, "dou" / "Do" → $\partial$, "x q" / "XQ" → $x^3$, "sine 1" → $\sin y$, "topol o topography" → topography, "extending to" → $x$ tending to.

## Coverage

| Source | Citations | Slides cited (slide × citations) |
|---|---|---|
| S001 | 56 | 4×1, 5×1, 6×1, 7×1, 8×4, 9×8, 10×5, 11×8, 12×5, 13×3, 14×2, 15×6, 16×3, 17×8 |
| CW | 5 | |
| ESA | 1 | |
| T001a | 39 | |
| T001b | 44 | |

Slides 001·1, 001·2 and 001·18 are title, session and closing slides. Slides 001·3–7 are motivation and are summarised in topic 1. The lecture announces the geometric interpretation (T001b 00:20:38); it is class 002.